

Partisan Differences in Natural Hazard Risk

Evidence from Registered Voters in Florida

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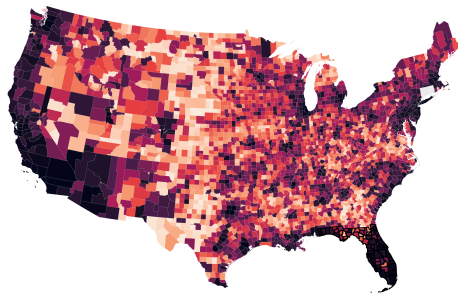
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CREEA Annual Meeting
2026

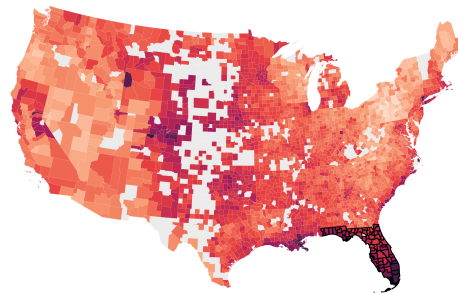
Natural hazard risk and homeowner insurance premiums across US counties



Expected Annual Loss (NRI percentile)

0 25 50 75 100

County expected annual loss from natural hazards. Source: FEMA National Risk Index (NRI)

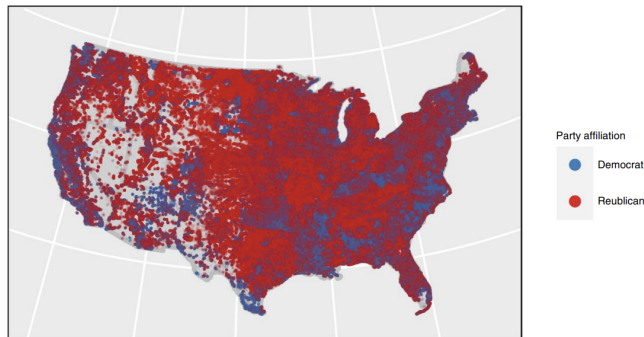


Mean annual premium (USD, log scale)

\$2,000.00 \$3,000.00 \$5,000.00

Mean annual homeowner premium by county. Source: Keys and Mulder (2024)

Partisan residential segregation in the US



Red = Republican (conservative)

Blue = Democrat (liberal)

US political colors are the reverse of Canada's

Partisan segregation of 180 million registered voters. Source: Brown and Enos (2021)

Literature: migration toward hazard and partisan risk beliefs

- ▶ Subsidized flood insurance draws migration into flood-prone communities (Peralta and Scott 2024)
- ▶ Compared to Republicans, Democrats are:
 - ▶ More likely to believe that climate change is happening and human-caused (McCright and Dunlap 2011)
 - ▶ More likely to believe that climate change will raise flood risk (Botzen et al. 2016)
 - ▶ More likely to evacuate ahead of a hurricane (Long et al. 2020)
 - ▶ Willing to pay more for coastal homes with lower flood risk (Bakkensen and Barrage 2022)
- ▶ Homes exposed to sea-level rise are increasingly owned by Republicans (Bernstein et al. 2022)
- ▶ Not yet measured for a whole electorate: whether political ideology predicts the hazard risk of migration [Contributions](#)

Florida: hurricanes and flooding

Coastal flooding

Local / State

Swept away: Aerial look at Helene's impact on Florida's Big Bend

WUSF | By [Diego Perdomo](#)

Published September 28, 2024 at 12:05 PM EDT



WUSF, 2024

Riverine flooding



NBC News, 2022

Hurricane



PBS News (AP), 2024

Two concurrent patterns in Florida

Migration into hazard-exposed areas; insurance costs rising



By —
Terry
Spencer,
Associated
Press

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What makes Tampa especially vulnerable to hurricanes right now

Nation Oct 7, 2024 6:01 PM EDT

PBS News (AP), 2024

YaleEnvironment360

Published at the Yale School of the Environment

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Emergency workers on Harbor Island, Florida, in the aftermath of Hurricane Ian, September 2022. © iStockphoto.com / AP Photo

How Climate Risks Are Putting Home Insurance Out of Reach

After years underestimating the risks posed by climate-fueled disasters, the U.S. home insurance industry is in turmoil. In vulnerable areas, rising insurance costs are sparking housing market and investment, as homeowners scramble to try to find insurance they can afford.

By SILBERT H. GUEL - SEPTEMBER 16, 2024

Yale Environment 360, 2025

Florida is turning Republican

NEWS / THE GAZZ ON FLORIDA POLITICS

Republicans make Florida history, pass Democrats in voter registration

Gov. DeSantis said on Friday morning that the GOP had struck the milestone they'd been approaching since 2012.

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Tampa Bay Times, 2021

Politics

Miami-Dade becomes the latest Florida county to flip from blue to red in voter registration

WFSN Public Media | By Mitch Perry, Florida Phoenix
Published May 26, 2025 at 1:29 AM EDT

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WUSF (WLRN), 2025

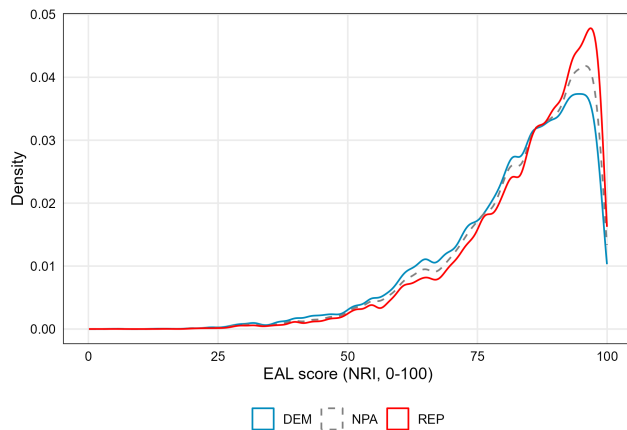
Research questions and designs

Research question	Design	Identifying comparison
Q1. Does ideology predict the hazard risk where a voter lives?	Cross-sectional comparison	Voters of the same county, sex, race and age, differing in party registration - observational
Q2. When voters migrate, does ideology predict the hazard risk they move to?	Difference-in-levels design	Comparable movers leaving the same county in the same window, from comparable origin tracts, differing in political ideology
Q3. Does voters' ideological realignment predict their change in hazard risk from migration?	Individual first-difference design	The same voter's ideology and risk exposure before and after the move
Mechanism	Decomposition of the Q2 gap across destination attributes	

Data

- ▶ Three snapshots of Florida's statewide voter registration file: February 2017, June 2022, November 2024
 - ▶ Party registration, residence address, sex, race, date of birth, voter ID; 13.5 to 16.0 million geocoded voters per snapshot
 - ▶ Linked by voter ID: 5.6 million moves between census tracts within Florida
- ▶ Each residence geocoded to its census tract (98-99% of records)
- ▶ Hazard risk: the tract's expected annual loss (EAL) score, from the FEMA National Risk Index
 - ▶ Percentile rank among US census tracts, 0 to 100
 - ▶ One overall score, and separate scores for coastal flooding, riverine flooding and hurricane [Maps](#)
 - ▶ Fixed across snapshots: a voter's exposure changes only through migration
- ▶ Tract income, home value, schooling, urbanicity (American Community Survey); distance to the coast and floodplain status of each address

Data: tract EAL score by party registration, 2017



Density of tract EAL score by party registration, 2017 snapshot. NPA: no party affiliation

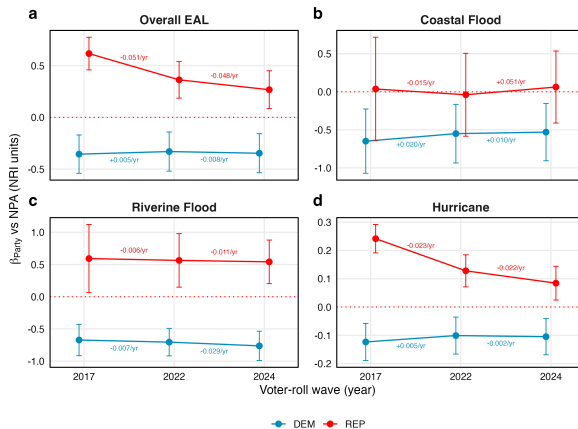
Q1: Does ideology predict the hazard risk where a voter lives?

- ▶ Voter i lives in census tract j of county c at snapshot t ; separately for each snapshot and each hazard, we estimate:

$$h_{jc} = \beta_D D_{ijct} + \beta_R R_{ijct} + X_i' \gamma + \mu_c + \varepsilon_{ijct}$$

- ▶ h_{jc} : EAL score of tract j in county c
- ▶ D_{ijct} , R_{ijct} : Democratic and Republican registration of voter i ; unaffiliated voters (NPA) omitted
- ▶ X_i : sex, race and age; μ_c : county fixed effect
- ▶ Identifying comparison: two voters of the same county, sex, race and age, differing in party registration - observational
- ▶ $\beta_R - \beta_D$: the within-county Republican-Democrat gap

Q1 results: party registration and EAL score, within county



Between and within counties

Sensitivity bounds

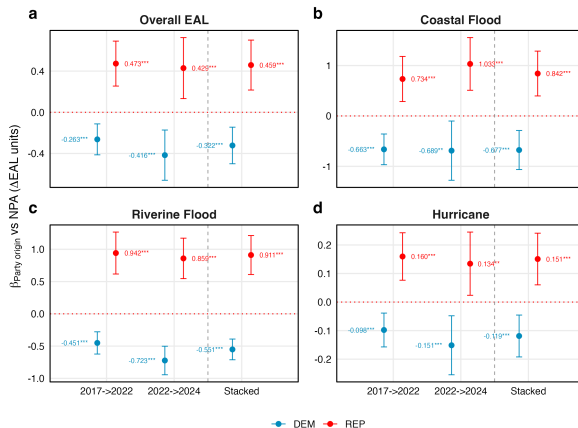
Q2: When voters migrate, does ideology predict the hazard risk they move to?

- ▶ Mover i leaves origin tract o in county c for destination tract d in county c' during window e (2017-2022 or 2022-2024); we estimate:

$$\Delta h_{ie} = \beta_D D_{ioce} + \beta_R R_{ioce} + X_i' \gamma + A_{oc}' \rho + \lambda_e + \mu_c + \varepsilon_{ie}$$

- ▶ $\Delta h_{ie} = h_{dc'} - h_{oc}$: destination EAL score minus origin EAL score
- ▶ D_{ioce}, R_{ioce} : Democratic and Republican registration at the start of window e , before the move; unaffiliated movers (NPA) omitted
- ▶ A_{oc} : origin-tract income, log home value, share with a bachelor's degree, urbanicity, log distance to the coast, floodplain status Control ladder
- ▶ X_i : sex, race and age; λ_e, μ_c : window and origin-county fixed effects
- ▶ Identifying comparison: movers leaving the same county in the same window, alike in sex, race, age and origin tract, differing in pre-move party registration
- ▶ $\beta_R - \beta_D$: the Republican-Democrat gap

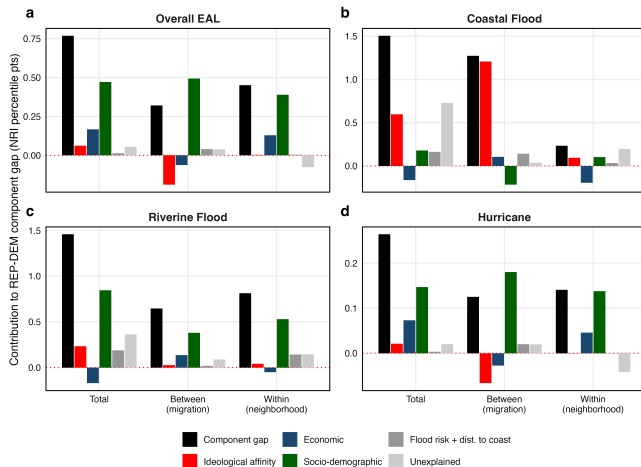
Q2 results: pre-move party registration and the change in EAL score



Between and within counties

Q2 mechanism: the Republican-Democrat gap, between and within counties

- ▶ Republican-Democrat gap split into its between-county and within-county parts, each allocated across four groups of destination attributes
- ▶ Between counties: political composition (red bars) accounts for the coastal part, 1.21 of 1.27 points
- ▶ Within counties: political composition contributes at most 0.09 points, in any hazard
- ▶ Schooling and urbanicity (green bars) account for the riverine gap at both scales: 0.38 of 0.65 between, 0.53 of 0.81 within



Coastal and riverine, between county

No re-sorting within counties

Q3: Does voters' ideological realignment predict their change in hazard risk from migration?

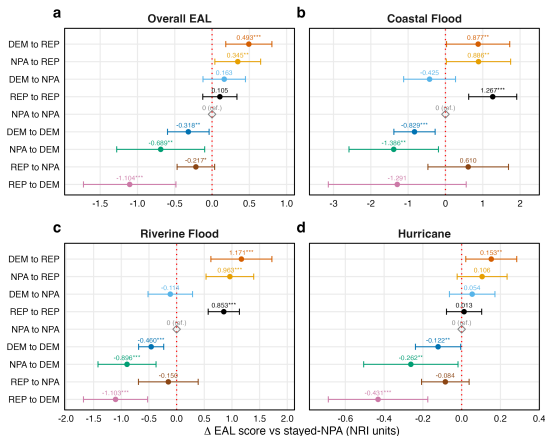
- ▶ Same movers, tracts and counties as Q2; we replace pre-move party registration with the nine registration pairs a mover can hold at the two dates:

$$\Delta h_{ie} = \sum_{q \neq N \rightarrow N} \theta_q \mathbf{1}\{Q_{ie} = q\} + X_i' \gamma + \lambda_e + \mu_c + \varepsilon_{ie}$$

- ▶ Q_{ie} : the pair (registration before the move, registration after); movers unaffiliated at both dates omitted
- ▶ θ_q : change in EAL score for movers on transition q , relative to movers who stayed unaffiliated
- ▶ Identifying comparison: the same voter's exposure before and after, so the starting position is netted out
- ▶ The file dates neither the address change nor the registration change within a window

Sequencing check

Q3 results: change in EAL score from migration, by ideological realignment



Senior and working-age movers

Conclusion

- ▶ Republicans live in riskier tracts than Democrats
 - ▶ Within-county gap: 0.97 points in 2017, 0.69 in 2022, 0.61 in 2024
- ▶ Political ideology predicts differential hazard exposure from migration
 - ▶ Republican-Democrat gap: 0.78 points overall; 1.52 coastal, 1.46 riverine
 - ▶ About \$29-35 per mover per year in insurance premium range [Back-of-envelope](#)
 - ▶ Coastal sorting goes with destination counties' politics, riverine sorting with schooling and urbanicity
- ▶ Voters' ideological realignment tracks their change in hazard risk from migration
 - ▶ Republican-to-Democrat realignment: 1.10 points safer; Democrat-to-Republican: 0.49 points riskier

Thank you so much for listening

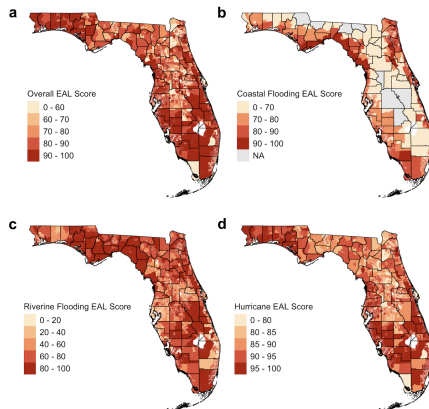
Comments? Questions?

Appendix: Contributions

- ▶ On partisan residential sorting (Brown and Enos 2021; Brown 2024; Brown et al. 2025)
 - ▶ We add the natural hazard content of that sorting: in levels, in moves, and within voters
- ▶ On climate beliefs and coastal housing markets (Bernstein et al. 2022; Bakkensen and Barrage 2022; Bakkensen and Ma 2020)
 - ▶ We measure realized exposure for a whole electorate, across four hazards, in the roll and in the moves
- ▶ On climate migration and the political economy of disasters (Hino et al. 2017; Sheldon and Zhan 2022; Healy and Malhotra 2009)
 - ▶ We add political ideology as a margin of climate migration and link the migration to a subsidized insurance pool

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Appendix: EAL scores by hazard, Florida census tracts

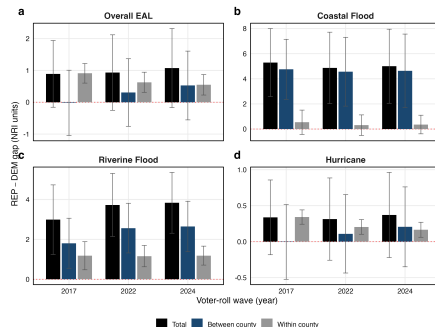


Tract EAL score: (a) overall, (b) coastal flooding, (c) riverine flooding, (d) hurricane.
County borders in black; grey = not applicable

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Appendix: The statewide gap, between and within counties

$$\text{gap}_t^{\text{statewide}} = \text{gap}_t^{\text{between-county}} + \text{gap}_t^{\text{within-county}}$$



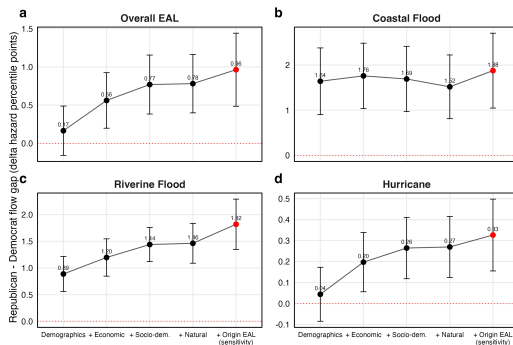
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Appendix: Omitted-variable bounds, spatial standard errors, permutation

- ▶ Oster (2019) bounds on the level estimates, demographics-only specification: Democratic versus unaffiliated survives ($\text{delta}^* = 3.70$); Republican versus unaffiliated does not ($\text{delta}^* = 0.85$)
- ▶ Cinelli and Hazlett (2020) robustness value about 0.002: a small partial R-squared, precisely estimated
- ▶ Conley (1999) spatial standard errors out to 200 km inflate the level SEs by about 70%; the Republican coefficient remains statistically significant
- ▶ Permutation of pre-move party registration within origin counties, 999 draws: randomization $P < 0.001$ for the Republican-Democrat gap among movers

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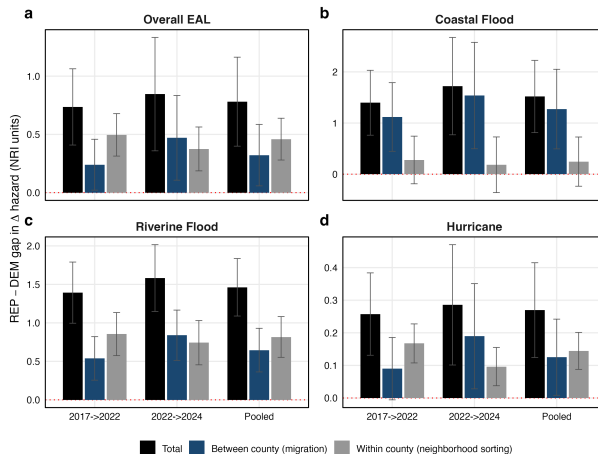
Appendix: Republican-Democrat gap as origin-tract controls are added



Republican-Democrat gap among movers, overall EAL: 0.17 with demographics only; 0.56 adding origin income and home value; 0.77 adding schooling and urbanicity; 0.78 adding coast distance and floodplain; 0.96 adding origin EAL score

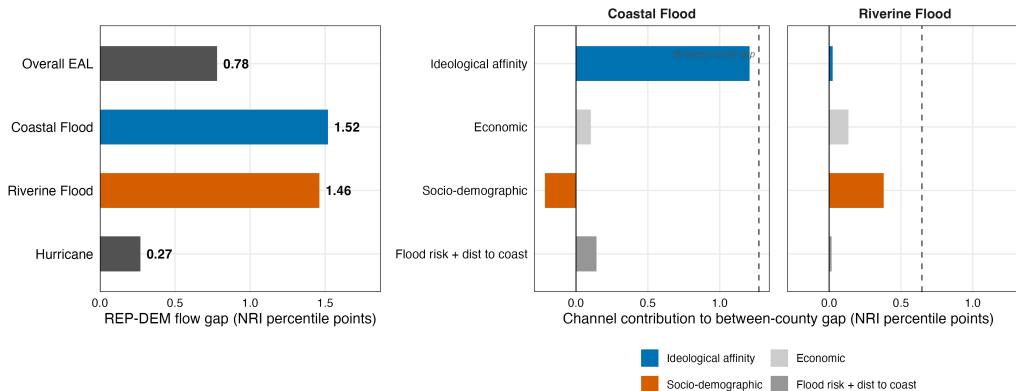
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Appendix: Movers' Republican-Democrat gap, between and within counties



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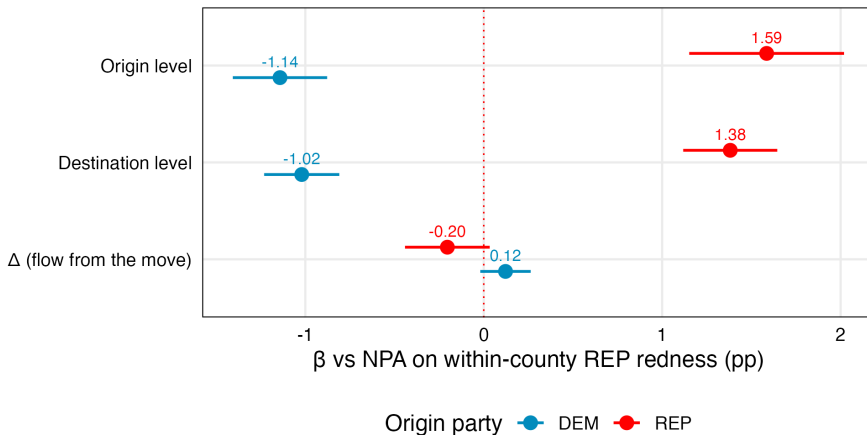
Appendix: Between-county part, coastal and riverine flooding



~95% of between gap; riverine Socio-demographic = 0.38 ~59% of between gap). Dashed line = between-county gap. NRI = FEMA National Risk Index. Note: the ~107%/~58% figures cited in the paper use the mediator-complete sample.

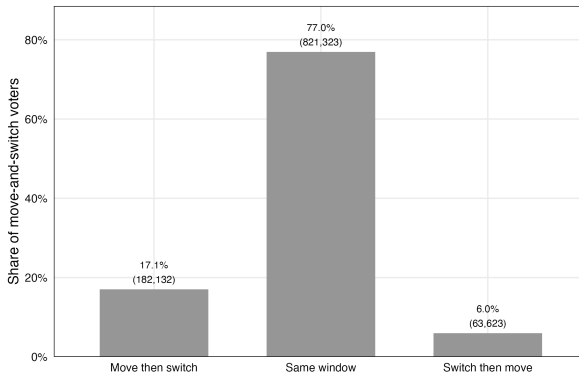
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Appendix: No political re-sorting within counties among movers



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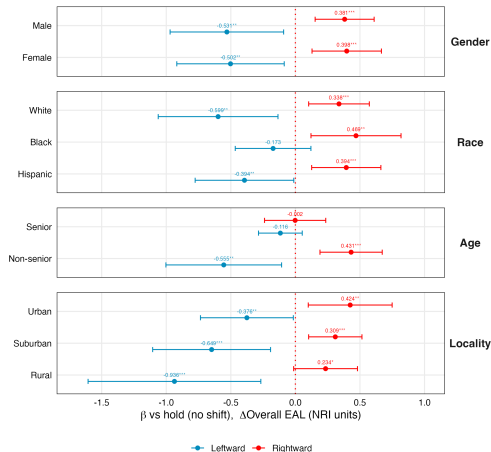
Appendix: Sequencing of moves and registration changes



Party registration in 2017 and in 2024, each predicting the 2017-2022 change in EAL score, same movers

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Appendix: Heterogeneity, senior and working-age movers



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Appendix: Back-of-envelope, the Republican-Democrat gap in insurance premiums

- ▶ County homeowner premiums rise with county EAL score (Keys and Mulder 2024): \$37-45 of annual premium per percentile point, in two specifications
- ▶ 0.78-point Republican-Democrat gap $\times$ gradient: about \$29-35 per mover per year in priced premium
- ▶ The added risk falls in the range that Citizens Property Insurance, Florida's insurer of last resort, underwrites below actuarial cost (Sastry, Sen and Tenekedjieva 2026)
- ▶ What is not paid in premium is carried by Citizens and, through its assessments, by policyholders statewide
 - ▶ The priced figure is thus a floor on what the movement costs others

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